Steak, Ramen, and Recessions: Other Elasticities

ECON 201: Principles of Microeconomics
Lecture 8

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Last Class: Price Elasticity of Demand

$$\varepsilon = -\frac{\%\text{ change in demand}}{\%\text{ change in price}}$$

- Last class we measured how much demand responds to a change in **price**.
- But price is not the only thing that moves demand. **Income** changes, and so do the prices of **other goods**.

Today's terms: the *income elasticity of demand*, *normal* and *inferior goods*, *necessities* and *luxuries*, the *cross-price elasticity of demand*, *substitutes* and *complements*.

Income Elasticity of Demand

The **income elasticity of demand** is the percentage change in demand that would occur in response to a 1% increase in income:

$$\varepsilon_{\text{income}} = \frac{\%\text{ change in demand}}{\%\text{ change in income}}$$

- There is no minus sign this time, because the sign is part of the answer.
- A positive elasticity means demand rises with income. A negative one means demand falls with income.

Worksheet, Activity 1

Maya works part time, and her monthly income rises from $1,000 to $1,200. Find the income elasticity for each good below.

$$\varepsilon_{\text{income}} = \frac{\%\text{ change in demand}}{\%\text{ change in income}}$$

	At $1,000	At $1,200	% change in demand	Income elasticity
Packs of ramen	40	32		
Gallons of gas	20	22		
Steak dinners	2	3		

Normal and Inferior Goods

- A **normal good** is one people buy more of as their income rises, so its income elasticity is positive.
- An **inferior good** is one people buy less of as their income rises, so its income elasticity is negative.
- Maya's income rose 20% and her ramen fell 20%, so $\varepsilon_{\text{income}} = -1$. Ramen is inferior for her.
- Her gas rose 10% and her steak dinners 50%, so both are normal, at 0.5 and 2.5.
- In US spending data, breads and cereals are inferior, with an income elasticity of -0.09.

Necessities and Luxuries

- Normal goods split in two, depending on whether the income elasticity is below or above 1.
- A **necessity** has an income elasticity between 0 and 1. People buy more of it as they get richer, but demand grows more slowly than income.
- A **luxury** has an income elasticity greater than 1. Demand for it grows faster than income.
- Gas is a necessity for Maya, at 0.5. Steak dinners are a luxury, at 2.5.

Income Elasticities in the United States

Estimates for the United States (Muhammad et al., 2011):

Category	Income elasticity
Food, beverages, and tobacco	0.35
Education	0.91
Housing	1.06
Transport and communications	1.13
Recreation	1.25

Below 1 is a necessity and above 1 a luxury. Housing counts as a luxury here, so the word is technical rather than everyday.

What Happens in a Recession

- In a recession, incomes fall. The goods with the highest income elasticities lose the most demand.
- Take a 10% fall in income. Spending on recreation falls by about 12.5%, and spending on food by about 3.5%.
- Demand for an inferior good moves the other way. It **rises** when incomes fall.
- As the 2008 recession set in, Nielsen recorded Spam sales up 10.6% over the twelve weeks to May 3, against the same weeks a year earlier.

Cross-Price Elasticity of Demand

The **cross-price elasticity of demand** for good A is the percentage change in demand for A that would occur in response to a 1% increase in the price of good B:

$$\varepsilon_{\text{cross}} = \frac{\%\text{ change in demand for A}}{\%\text{ change in the price of B}}$$

- Again there is no minus sign, because the sign is what tells us how the two goods are related.
- A rise in the price of one good sends some buyers toward another good, and away from a third.

Worksheet, Activity 2

The campus cafe raises the price of a coffee from $2.50 to $3.00. Find the cross-price elasticity for each good below.

$$\varepsilon_{\text{cross}} = \frac{\%\text{ change in demand for A}}{\%\text{ change in the price of B}}$$

	At $2.50	At $3.00	% change in demand	Cross-price elasticity
Coffee across the street	6	9		
Donuts at the campus cafe	8	6		
Packs of ramen	32	32		

Substitutes and Complements

- Two goods are **substitutes** if a rise in the price of one raises demand for the other, so $\varepsilon_{\text{cross}}$ is positive.
- Two goods are **complements** if a rise in the price of one lowers demand for the other, so $\varepsilon_{\text{cross}}$ is negative.
- The campus price rose 20% and coffee across the street rose 50%, so $\varepsilon_{\text{cross}} = 2.5$. The cafes are substitutes.
- Donuts fell 25%, so $\varepsilon_{\text{cross}} = -1.25$: coffee and donuts are complements. Ramen did not move, so at 0 it is neither.

Soda Just Outside Philadelphia

Philadelphia's tax raised the price of a sweetened drink inside the city by 34%. What happened at the stores just outside it?

- Sales at the average store inside the city fell by about 56,000 ounces a week.
- Sales at the average store within two miles outside the city rose by about 64,000 ounces a week.
- Soda inside the city and soda just outside it are close substitutes, so the cross-price elasticity between them is large and positive (Seiler, Tuchman, and Yao, 2021).

Sources

- Muhammad, Seale, Meade, and Regmi (2011), “International Evidence on Food Consumption Patterns: An Update Using 2005 International Comparison Program Data,” USDA Economic Research Service Technical Bulletin 1929, appendix tables 1 and 5.

- Associated Press (2008), “Sales of Spam rise as consumers trim spending,” NBC News, May 28, reporting Nielsen Company data.

- Seiler, Tuchman, and Yao (2021), “The Impact of Soda Taxes: Pass-Through, Tax Avoidance, and Nutritional Effects,” Journal of Marketing Research 58(1), table 5.